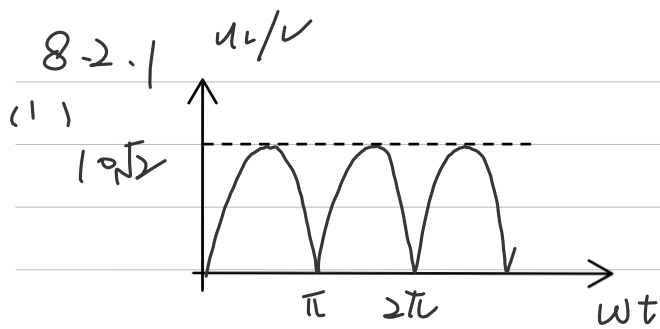




导电路径:

正半周: $a \rightarrow D_1 \rightarrow R_L \rightarrow o$.

负半周: $b \rightarrow D_2 \rightarrow R_L \rightarrow o$.

$$(2) U_L = \frac{1}{\pi} \int_0^{\pi} \sqrt{2} U_2 \sin \omega t d(\omega t) = 0.9 U_2 = 9 V.$$

$$(3) I_L = U_L / R_L = 9 / 100 = 0.09 A.$$

$$I_D = \frac{1}{2} I_L = 0.045 A.$$

$$(4) U_{DPM} = \sqrt{2} U_2 = 28.28 V.$$

8.2.2

$$I_L = U_L / R_L = 0.25 A \quad I_D = \frac{1}{2} I_L = 0.125 A$$

$$U_2 = 1.11 U_L = 33.3 V$$

$$U_{DPM} = \sqrt{2} U_2 = 47.1 V.$$

$$S = U_2 I_2 = 9.24 VA.$$

8.2.3

$$(1) T = 0.02 S \quad \beta_{LC} = 0.04 S > (3 \sim 5) \frac{T}{2}$$

$$U_0 \approx 1.2 U_2 = 24 V$$

$$(2) \text{电容充电: } U_0 \approx U_{2-\max} = \sqrt{2} U_2 = 28.28 V$$

$$(3) \text{整流后无滤波: } \therefore U_0 = 0.9 U_2 = 18 V.$$

$$(4) \text{半整流波无滤波.}$$

$$U_0 = 0.45 U_2 = 9 V.$$

8.2.6.

R_p 最下端时. $U_0 \rightarrow \max.$

$$U_{0\max} = \frac{1}{R_2} (R_1 + R_2 + R_p) (U_2 + U_{BE2}) = 16 V$$

R_p 最上端时. $U_0 \rightarrow \min.$

$$U_{0\min} = \frac{R_1 + R_p + R_2}{R_2 + R_p} (U_2 + U_{BE2}) = 5.3 V.$$

